

eties to improve ethanol yield. For years, maize prices hovered around ₹15.9 per kg, but over the past three years, they have surged to ₹24.6 per kg, a sharp 50% increase, reflecting the shift in demand dynamics.

Even with this expansion, current production is still short of projected needs. According to government data, India imported eight lakh tonnes maize in 2024-25, sharply up from 1.27 lakh tonnes the previous year, to meet ethanol demand. Demand is projected to reach 650-700 lakh tonnes per year to meet the E30 blending target by 2030.

SOPS FOR ADOPTION

Despite the ambitions to reduce the fuel import bill and provide better returns to farmers, car owners have flagged a series of concerns that need to be addressed. At present, India does not have flex-fuel vehicles, though the government has issued a mandate to OEMs to ramp up production of vehicles that can run on any blend up to E100. “Almost 19 flex fuel cars were showcased at the Bharat Mobility Expo 2025 but the OEMs are waiting for a conducive policy by the government,” says Ballani.

However, a flex-fuel vehicle will cost more. “The shift to flex-fuel technology isn’t without cost implications. Upgrading the powertrain and related components adds around ₹60,000 or more to the price of a model like Wagon R. The key question is: who bears this additional cost?” says S&P’s Sharma.

All stakeholders agree that the government needs to offer incentives for flex-fuel vehicles and treat them on a par with electric vehicles, which are taxed at 5% under the goods and services tax (GST). Sharma believes that unless targeted incentives are introduced—such as GST reduction,

THE MAIZE PUSH

- Maize sowing area has grown by over 8.74%, the largest growth in any crop this kharif
- Ethanol programme has helped maize farmers with a 50% price jump in five years

YEAR	MAIZE PRICE (₹/TONNE)
2021	15,920
2022	21,095
2023	22,247
2024	23,500
2025*	24,632

NOTE *2025 IS TILL JULY
SOURCE CLFMA OF INDIA



CEREAL GRAINS LIKE MAIZE AND DAMAGED FOOD GRAINS —INCLUDING BROKEN RICE—NOW ACCOUNT FOR ROUGHLY 52.7% OF INDIA'S ETHANOL OUTPUT

policy support around taxation, or adjustments to CAFE norms — consumers are unlikely to adopt this technology. “The complication, however, is that flex-fuel vehicles can still run entirely on gasoline, which makes designing such incentives far more complex,” he says. Car owners have also flagged concerns around insurance claims. Most standard motor insurance policies in India exclude coverage for damage resulting from the use of incorrect fuel. As a result, if a non-E20-compliant vehicle sustains engine damage due to the use of E20 fuel, the insurer may reject or reduce the claim. The Society of Indian Automobile Manufacturers, an auto industry body, has said that E20 fuel-related insurance and warranty claims will be honoured.

But insurance companies have a different take. “In case of an engine failure due to incorrect fuel usage, the claim would not be admissible. This falls under gross negligence as per the policy terms,” ACKO Insurance, a major general insurance company, has clarified on customers’ social media queries.

WHAT NEXT?

The government has indicated that any shift toward E27 will be contingent upon detailed discussions with all stakeholders. While trials for higher blends are under way, scientists caution that moving beyond E20 will require significant adjustments—including retrofitting existing vehicles and promoting flex-fuel technology.

As E20 continues to be dispensed nationwide, the responsibility of addressing any vehicle-related implications rests with the consumer, at least until further safeguards or standards are introduced. **BT**

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